

Connecting NCE Cab Bus to MAX485

<https://gemini.google.com/app/c211e6c28c9a9d8f>

User prompt: connect NCE Cab Bus to Max485

Response: Connecting an **NCE Cab Bus** to a **MAX485 module** allows a microcontroller (like an Arduino, ESP32, or STM32) to communicate with NCE Digital Command Control (DCC) equipment.

The NCE Cab Bus operates as an RS-485 serial network running at **9600 baud, 8 data bits, no parity, and 2 stop bits (8N2)**.

1. NCE Cab Bus Pinout (RJ12 6P6C)

Looking at the front of a standard NCE UTP panel or Cab Bus RJ12 socket (tab facing down, pins 1 to 6 left to right):

Pin	NCE Wire Function	Description
Pin 1	White	Ground (GND)
Pin 2	Black	Ground (GND)
Pin 3	Red	RS-485 Line A (+)
Pin 4	Green	RS-485 Line B (-)
Pin 5	Yellow	+12V DC (Power for Cabs)
Pin 6	Blue	+12V DC (Power for Cabs)

2. MAX485 to NCE Cab Bus Wiring

Connect the **RS-485 bus side** of the MAX485 module directly to the RJ12 Cab Bus cable:

MAX485 Pin/Terminal	NCE Cab Bus Connection	Notes
A	Pin 3 (Red)	RS-485 Non-Inverting Line (+)
B	Pin 4 (Green)	RS-485 Inverting Line (-)
GND	Pin 1 or 2 (White/Black)	Shared Common Ground

- **Power Warning:** Do **NOT** connect Pin 5 or 6 (+12V DC) directly to your MAX485 module or microcontroller logic lines. Supply your MAX485 with standard **+5V DC** from your microcontroller or an external 5V regulator.

3. MAX485 to Microcontroller Wiring

MAX485 Pin	Microcontroller Connection	Function
VCC	5V	5V Power Supply
GND	GND	Common Ground
RO (Receiver Output)	RX (e.g., Arduino Pin 0 or Serial RX)	Sends received data to MCU
DI (Driver Input)	TX (e.g., Arduino Pin 1 or Serial TX)	Sends MCU data to bus
DE (Driver Enable)	Digital GPIO Pin (e.g., Pin 2)	HIGH to Transmit
RE (Receiver Enable)	Digital GPIO Pin (Tied with DE)	LOW to Receive (Active Low)

Tip: Tie **DE** and **RE** together on the MAX485 module. Set the GPIO pin **LOW** by default to listen for pings from the NCE Command Station. Set the pin **HIGH** only when transmitting a response.

4. Code & Protocol Setup Essentials

- **Serial Configuration:** Set your HardwareSerial port to 9600 Baud with 2 Stop Bits:

```
// Arduino Hardware Serial initialization for NCE Cab Bus
Serial.begin(9600, SERIAL_8N2);
```

- **Bus Addressing & Polling:** The NCE Command Station acts as the master, sequentially polling cab addresses (0x80 broadcast ping followed by cab address). Your microcontroller must wait to be polled before asserting DE/RE HIGH to send data.
- **Libraries:** Consider using open-source MRRWA libraries (**MRRWA** / **NceCabBus**) which simplify parsing cab commands, throttle inputs, and AIU (Auxiliary Input Unit) emulation.

Sources

1. [MAX485 IC: RS-485 Communication Driver -Pinout, Datasheet](#)
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